BTS-IV-04.19- 0309 Reg.No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

A

B.Tech. Degree IV Semester Examination April 2019

CS/EB 1402 MICROPROCESSORS

(2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A

(Answer *ALL* questions)

(8× 5 = 40)

I. (a) Write a note on how serial communication is implemented in Microprocessor 8085.

(b) Discuss the different microprocessor families.

(c) Compare memory mapped I/O and I/O mapped I/O.

(d) Write the usages of various shift/rotate instructions in Microprocessor 8085.

(e) What is meant by Interrupt Service routine and Interrupt vector?

(f) What is the usage of NOP mnemonic in Microprocessor 8085?

(g) Write a note on how memory Chip interfaces with 8085.

(h) Write the Control word when PPI 8255 works in simple I/O mode with Port A output mode, Port B – Input mode, Port C bit 1 – Input mode, Port C bit 6 Output mode.

PART B

(4 × 15 = 60)

II. (a) Explain the functional details of all Pins in Microprocessor 8085. (10)

(b) Write a note on DMA features. (5)

OR

III. (a) Explain the register organization in Microprocessor 8085. (8)

(b) Explain Bus Demultiplexing in Microprocessor 8085 using diagram. (7)

IV. (a) Write an Assembly Language program in Microprocessor 8085 to find the largest number in a set of numbers stored from memory location 2001 onwards. The count of numbers stored in location 2000 and store the largest number in the following location. (10)

(b) Write a note on logical instructions available in 8085 Microprocessor. (5)

OR

V. (a) Write an Assembly Language program in Microprocessor 8085 to find the number of ones in a set of numbers stored from memory location 2001 onwards. The count of numbers stored in location 2000 and store the number of ones in each number in the corresponding location 3001 onwards. Eg: $(2001) = 73H$ $(3001) = 05h$ (10)

(b) Write the usages of any Five instructions in implicit mode. (5)

(P.T.O.)

VI.	(a)	Draw the Timing Diagram of any one instruction having 13 T states.	(8)
	(b)	Write a note on Restart Instructions in Microprocessor 8085.	(7)
		OR	
VII.	(a)	Write the detailed procedure of how the Microprocessor handles a Non-Vectored Interrupt.	(8)
	(b)	Why branch instructions in Microprocessor takes different number of T states in true and false conditions?	(7)
VIII.	(a)	Explain the functional details of all pins in peripheral chip 8279.	(8)
	(b)	Explain the block diagram and working of peripheral chip 8253.	(7)
		OR	
IX.	(a)	Explain the block diagram and working of peripheral chip 8251.	(8)
	(b)	How the PPI 8255 can be interface with Microprocessor 8085 in memory mapped I/O?	(7)

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